

PRODUCT REFERENCE / EXPERIENCE SPECIFICATION

OMNILUME

COMPLETE FEATURE GUIDE

Scope	Existing, partial, missing, blocked, unverified, planned, and newly requested capabilities.
Boundaries	Rooms are separate from Personal Chat; groups are explicit conversions; AI is last.
Evidence	Every feature follows source/static -> local runtime -> multi-user/browser -> database/security -> production.
Status	Implemented in source is not the same as fully verified.

Protect the existing codebase. Extend first. Verify source, runtime, security, multi-user, and production behavior separately.

How to use this guide

Each feature is written as a product and engineering contract. It says what the user should experience, what the current code provides, what must be added, what security boundary applies, what it depends on, and how completion is proven.

Non-negotiable product boundaries

- Rooms are collaborative spaces with room membership, Watch, Study, Files, presence, roles, notifications, and lifecycle.
- Personal Chat is separate. Personal contains accepted friends and persistent groups. General contains non-friend conversations and requests.
- A room becomes a group only through authorized transactional conversion. The group keeps the room name but loses the room username/public link.
- Room Memories are normal saved human moments/history. Room Memory is a later AI retrieval/index feature.

Canonical new-feature coverage

The exact names below are the 54 newly requested feature requirements. Each is integrated into the corrected action sequence and covered by the detailed domain entries that follow.

Feature	Owning action	Status
Edit/delete own messages	08	Partial
Shared PDFs	10/12	Partial
Invite link + unique code	05/09	Missing
Live shared notes	12	Missing
Timer-created owner notification	12	Missing
Member 3-dot actions	08	Missing
Ghost Mode	05/09	Missing
Voice chat + owner mute	12	Partial
Room Control Center	05/09	Missing
Room Roles & Permissions	05/09	Missing
Room Announcements	08	Missing
Pinned Messages	08	Missing
Message Replies / Threads	08	Missing
Mentions	08	Missing
Notification Center	08	Missing
Room Search	13	Partial
Activity / Room History	13	Missing
Custom Room Layout Presets	05/09	Missing
Owner-controlled room UI/layout	05/09	Missing
Room Themes / Appearance	05/09	Missing
Polls & Voting	12	Missing
Shared Queue	11	Missing
Now Playing / Shared State Bar	11	Missing

Feature	Owning action	Status
User Presence States	05/09	Missing
Leave / Kick / Ban / Block	05/09	Missing
Screen Sharing	12	Missing
Raise Hand	12	Missing
Room Rules & Welcome Message	05/09	Missing
Temporary Guest Access	05/09	Missing
Room Lock	05/09	Missing
Scheduled Rooms	12	Missing
Room Events	12	Missing
Shared Whiteboard Cursor Presence	10/12	Missing
Collaborative Document Editing	10/12	Missing
Room Bookmarks / Saved Content	05/09	Missing
Room-Specific User Profiles	05/09	Missing
Smart Room Search & Filters	13	Missing
Room Backup / Export	13	Missing
Shared Countdown / Event Timer	10/12	Missing
Live Activity Bar	13	Missing
Room Dashboard / Home	13	Missing
Room Clips / Highlights	11	Missing
Smart Auto-Organization	13	Missing
Room Templates	13	Missing
Guest / Temporary Identity Mode	05/09	Missing
Room Reputation / Trust Levels	13	Missing
Smart Notification Controls / Do Not Disturb	08	Missing
Cross-Room Personal Dashboard	13	Missing
AI Room Assistant	17	Planned
Live AI Meeting Assistant	05/09	Missing
AI-assisted Action Items / Tasks	17	Planned
Automated Room Summary	17	Planned
Meeting Minutes / Meeting Summary	17	Planned
Room Memory	17	Planned

Identity & Social

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Google OAuth and first-run identity

Partial

Identity & Social

What, why and experience

Google is the current target entry point; existing password login needs an explicit compatibility decision.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Auth clients, proxy, and profile records exist; add callback, setup gate, safe redirects, and provider configuration.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: RLS protects identities and private fields; no duplicate profiles or token leakage.

Dependencies: Actions 01-03

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Account switching and recovery/password policy

Missing

Identity & Social

What, why and experience

Switch supported identities without mixing sessions; document future password/recovery scope.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add isolated account selector and recovery decision without deleting existing users.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Never cross-contaminate sessions, keys, chats, or provider accounts.

Dependencies: Action 02

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Profiles, usernames, DOB and age handling

Partial Identity & Social

What, why and experience

Collect validated name, DOB, gender, avatar, bio, lowercase username, and visibility.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Extend profiles additively; derive age-sensitive rules server-side and minimize DOB exposure.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Protect DOB/gender and use privacy-safe defaults.

Dependencies: Actions 02-04

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Public/private profiles, posts, followers and following

Missing Identity & Social

What, why and experience

Public profiles are discoverable; private posts/lists require accepted followers or friends.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add relationship tables, posts, request states, and visibility RLS.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Private requests/content are owner/accepted relationship scoped.

Dependencies: Action 04

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Friends and relationship actions

Missing

Identity & Social

What, why and experience

Friend requests are separate from following; acceptance creates mutual follows safely.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Normalized friendship keys, recipient-only decisions, no duplicates/self requests.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: No public friend counts or unauthorized request reads.

Dependencies: Action 04

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Rooms & Permissions

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Room creation, joining, invite link and unique code

Partial Rooms & Permissions

What, why and experience

Authenticated users create and join by supported ID, username, and link formats.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Preserve createRoom/processRoomJoin; add safe invite lifecycle.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: RLS, approved membership, privacy, active lifecycle.

Dependencies: Actions 02-06

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room Control Center

Missing

Rooms & Permissions

What, why and experience

One place for owners/admins to manage room state and permissions.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Extend existing settings/member actions with granular controls.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Server/RLS enforcement, not hidden buttons.

Dependencies: Actions 05-06

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room Roles & Permissions

Partial

Rooms & Permissions

What, why and experience

Owner/admin/member semantics exist; granular permissions need expansion.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Permission matrix, transfer, immutable membership fields.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Least privilege and owner protection.

Dependencies: Actions 03,05

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room leave, kick, ban and block

Partial

Rooms & Permissions

What, why and experience

Leaving should clear membership, presence, subscriptions, and local state.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Add explicit transitions and access revocation.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Former/pending/rejected/banned states denied.

Dependencies: Actions 05-07

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Ghost Mode / anonymous room identity

Missing

Rooms & Permissions

What, why and experience

Use a controlled room display identity without exposing unnecessary profile details.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add scoped alias with moderation/audit visibility.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Not an authorization or accountability bypass.

Dependencies: Actions 04-07

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room rules, welcome, lock, themes and owner-controlled layout

Missing

Rooms & Permissions

What, why and experience

Owners set expectations and configure safe room presentation.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Persist settings/features and enforce in actions/RLS/realtime.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Private metadata remains private.

Dependencies: Actions 05-09

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Chat & Notifications

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Edit/delete own messages

Partial

Chat & Notifications

What, why and experience

A sender edits or deletes only their own eligible message.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Extend existing message actions with update ownership and immutable scope.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Current member, active room, sender, RLS/trigger.

Dependencies: Action 08

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Replies / Threads, reactions, GIFs and mentions

Missing

Chat & Notifications

What, why and experience

Messages can carry context, reactions, media, and targeted mentions.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add metadata/relations and notification scoping.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: No raw ciphertext/plaintext leakage; mention privacy.

Dependencies: Actions 07-08

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Announcements, pinned messages and Notification Center

Missing

Chat & Notifications

What, why and experience

Important room information is persistent and privately delivered.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add typed notifications, pins, read state, and authorized announcement actions.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Unrelated users receive nothing.

Dependencies: Action 08

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Smart notification controls / DND

Missing

Chat & Notifications

What, why and experience

Mute categories for a duration or forever, with mention preferences.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Per-user/per-room preferences and server filtering.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Do not suppress security/access notifications improperly.

Dependencies: Action 08

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Personal, General and Group Messaging

Missing

Chat & Notifications

What, why and experience

Friends/groups live in Personal; non-friends and requests live in General.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add conversation types, requests, group messages, read state, reclassification.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: RLS by friendship, follow/request, or group membership.

Dependencies: Action 15

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

E2EE private chat and undecryptable state

Partial

Chat & Notifications

What, why and experience

Keys stay on devices; one bad message does not break the list.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Preserve allSettled UI and document key/re-key/multi-device limits.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Never send keys/plaintext or sensitive crypto errors.

Dependencies: Actions 03,15

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Files & Study

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Shared PDFs with room-wide visibility and realtime notification

Partial

Files & Study

What, why and experience

Members see protected PDFs and are notified of shared availability.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Extend Files/storage metadata and typed notification event.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Room membership, active lifecycle, actual object path.

Dependencies: Actions 07,10

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Permanent files and temporary direct links

Missing

Files & Study

What, why and experience

Files can persist or be shared through bounded private links.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add metadata, signed URLs, retention and cleanup.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: No public storage bypass; links expire.

Dependencies: Actions 03,10

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Live shared notes and collaborative documents

Missing

Files & Study

What, why and experience

Members edit common notes/documents with reconnect recovery.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Versioned records, conflict handling, snapshots.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Member-scoped reads/writes and retention.

Dependencies: Actions 07,12

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Study timer, history, music and lecture media

Partial

Files & Study

What, why and experience

Existing Study/timer expands to history, music and supported educational video.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Extend StudyStage/session records and media references.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Focus Lock/lifecycle preserved.

Dependencies: Actions 06,10,12

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Whiteboard cursor presence, polls and shared countdown

Missing

Files & Study

What, why and experience

Members collaborate visually and coordinate decisions/time.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Authorized room events/state records with reconnect handling.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Private room activity only.

Dependencies: Actions 07,12

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Watch & Media

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Watch playback and seek reliability

Partial

Watch & Media

What, why and experience

Multiple users stay synchronized on play/pause/seek.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Preserve React Player v3, currentTime, throttling, suppression, drift correction.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Media source and controls follow room/storage RLS.

Dependencies: Actions 07,10,11

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Subtitles, audio tracks, metadata, lyrics and fallback providers

Partial

Watch & Media

What, why and experience

Media shows useful metadata and honest unsupported states.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Native tracks plus provider-neutral metadata/cache/attribution/fallback.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: No false playback claims.

Dependencies: Actions 10-11

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Shared Queue, Now Playing and Room Clips / Highlights

Missing

Watch & Media

What, why and experience

Members share media context and save moments.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Persist authorized queue/state/clip refs in Watch, not as a forward Music dependency.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Control permissions and retention.

Dependencies: Action 11

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Local media casting and Screen Sharing

Partial

Watch & Media

What, why and experience

Users share supported media or screen by consent and permission.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Use browser APIs, consent, cleanup, and room state.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Never expose private local/provider content.

Dependencies: Actions 05,07,11,12

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Collaboration & Voice

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Voice chat with owner mute/unmute

Missing

Collaboration & Voice

What, why and experience

Members join room voice; authorized owner/admin can moderate.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Use approved WebRTC/SFU, consent, mute state, reconnect.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Room membership and participant privacy.

Dependencies: Actions 05-07,12

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Raise Hand and Live Activity Bar

Missing

Collaboration & Voice

What, why and experience

Members signal attention and see lightweight room activity.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Authorized ephemeral events plus aggregate activity.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Avoid silent recording/telemetry.

Dependencies: Actions 07,12,13

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room Events, Scheduled Rooms and Shared Countdown/Event Timer

Missing

Collaboration & Voice

What, why and experience

Rooms can be planned and reminders/status shared.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Schedules, time zones, reminders, lifecycle integration.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Private events and protected cron.

Dependencies: Actions 06,12,13

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Manual Tasks/Action Items, recording consent and transcription

Missing

Collaboration & Voice

What, why and experience

Normal tasks and searchable meeting records exist before AI.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add task, consent, recording/transcript retention and search.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Strict consent/deletion/membership rules.

Dependencies: Action 12 before 17

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Search, Events & Management

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Room Search and Smart Room Search & Filters

Partial

Search, Events & Management

What, why and experience

Find eligible public rooms with useful filters.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Index safe public fields and scope private results.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Never enumerate private rooms.

Dependencies: Action 13

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Activity / Room History, Room Pulse and Room Memories

Missing

Search, Events & Management

What, why and experience

Users see what happened and save ordinary human moments.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Typed activity/history, aggregate Pulse, saved-memory records.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Privacy-minimized retention; distinguish from AI Room Memory.

Dependencies: Action 13

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room Dashboard / Home and Cross-Room Personal Dashboard

Missing	Search, Events & Management
What, why and experience See room status and cross-room personal overview. Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear. User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.	Current and implementation Current status: Planned or missing; no implementation should be inferred. Reuse: Aggregates with membership-scoped queries. Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.
Security and dependencies Security: No cross-room leakage. Dependencies: Action 13	Impact, target and testing Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics. Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect. Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room Bookmarks / Saved Content, Templates and Smart Auto-Organization

Missing	Search, Events & Management
What, why and experience Save useful content and organize/start rooms consistently. Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear. User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.	Current and implementation Current status: Planned or missing; no implementation should be inferred. Reuse: Per-user bookmarks, owner templates, explainable organization. Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Bookmarks inherit source permission; templates copy no private data.

Dependencies: Action 13

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room Backup / Export and Trust Levels

Missing

Search, Events & Management

What, why and experience

Authorized users export data and manage trust signals.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Scoped export jobs, audit, trust/reputation, safe downloads.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Owner/admin only; no private keys.

Dependencies: Actions 05,13,18

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Moderation, reports, reputation, leave/kick/ban/block

Missing

Search, Events & Management

What, why and experience

Abuse controls protect people and rooms.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Audit trail, reports, bans, escalation, owner/platform boundary.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Least privilege and evidence privacy.

Dependencies: Actions 05,13,18

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Groups & Music

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Room-to-Group Conversion

Partial

Groups & Music

What, why and experience

Explicit authorized conversion creates one persistent group.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: The architecture exists only partially or as a foundation.

Reuse: Remove group username/link, preserve name/origin/members/content, idempotent transaction.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: RLS and no destroyed-content resurrection.

Dependencies: Action 14

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Personal Groups and Group Workspace

Missing

Groups & Music

What, why and experience

Groups are chat-first but can open a room-style workspace.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Group roles/members/settings, three-dot workspace, Back navigation.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Group membership RLS; no automatic room username.

Dependencies: Actions 09,14,15

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Personal Music uploads and player

Missing

Groups & Music

What, why and experience

Private uploaded music supports queue and metadata for 48 hours.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Multi-upload, player, expiry, metadata/artwork/lyrics.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Private storage and cleanup; no R2 assumption.

Dependencies: Action 16

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Spotify, Apple Music, TIDAL, Qobuz and YouTube Music connections

Missing

Groups & Music

What, why and experience

Connect multiple providers through official OAuth/SDK paths.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Server-side adapters, search, playlists, disconnect/revoke, explicit limits.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Tokens protected; no scraping/download bypass.

Dependencies: Action 16

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room and Group Music synchronization

Missing

Groups & Music

What, why and experience

Authorized shared track/state while credentials remain personal.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Safe references/state through shared provider with controls.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Provider rules may limit shared playback.

Dependencies: Actions 07,14-16

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

AI & Intelligence

This domain follows the corrected action order. The owning action is the single implementation home; later domains may reuse it without creating a parallel source of truth.

Lume Study, Lume Watch and Lume Music

Planned

AI & Intelligence

What, why and experience

Named AI surfaces build on normal study, media, and music records.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Add permission-aware adapters and source-linked outputs.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: No cross-room retrieval or provider token access.

Dependencies: Action 17

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

AI Room Assistant

Planned

AI & Intelligence

What, why and experience

Answers room questions using permitted context.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Scoped retrieval, citations, consent, rate limits.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Recheck membership at query time.

Dependencies: Action 17

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Live AI Meeting Assistant

Planned

AI & Intelligence

What, why and experience

Consent-based assistance during live meetings.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Recording/transcription controls, stop, retention, participant consent.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: No silent recording or broad broadcast.

Dependencies: Actions 12,17

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

AI-assisted Action Items / Tasks

Planned

AI & Intelligence

What, why and experience

Drafts tasks from verified records for user approval.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Use manual task model and source references.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: No automatic permission-changing actions.

Dependencies: Actions 12,17

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Automated Room Summary and Meeting Minutes / Meeting Summary

Planned

AI & Intelligence

What, why and experience

Creates drafts from notes/history/transcripts.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Require source attribution, approval, correction, deletion.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Expired/deleted data is not resurrected.

Dependencies: Actions 08,12,17

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Room Memory

Planned

AI & Intelligence

What, why and experience

AI retrieval over approved room history; separate from ordinary Room Memories.

Why OmniLume needs it: It adds useful, privacy-aware collaboration while keeping room and personal boundaries clear.

User experience: the user sees clear controls and honest loading, success, error, empty, retry, and accessibility states.

Current and implementation

Current status: Planned or missing; no implementation should be inferred.

Reuse: Scoped index, retention, deletion, membership-aware retrieval.

Implementation: add the smallest compatible route/component/action/schema seam; preserve existing happy paths.

Security and dependencies

Security: Private data and expiry boundaries remain active.

Dependencies: Actions 06,13,17

Impact, target and testing

Impact: adds capability without changing unrelated room, E2EE, lifecycle, or session semantics.

Target: safe, retryable, accessible, privacy-preserving, and consistent after refresh/reconnect.

Testing: source/static -> local runtime -> independent multi-user/browser -> database/security -> production.

Definition of Done

- Happy path and important failures are actually observed, not assumed.
- Permission boundaries pass direct calls and UI tests.
- Retention, deletion, rollback/disable behavior, and safe logging are documented.

Existing foundation index

These systems remain protected foundation components while later actions extend them.

Foundation	Status	What is preserved
Supabase SSR proxy/session refresh	Partial/Unverified	src/proxy.ts protects routes and refreshes sessions; true expiry runtime evidence remains a gate.
RLS lockdown/storage/lifecycle migrations	Partial/Unverified	Migrations 002-005 and policies define room/member/message/storage/lifecycle boundaries; remote evidence must stay current.
RoomRealtimeProvider/useRoomSync	Partial/Unverified	One room synchronization architecture; reconnect/late-join breadth remains a gate.
Room chat/Delete for Everyone	Partial/Unverified	Current server/RLS ownership and propagation remain; richer actions layer on top.
E2EE private chat	Partial	Device-side keys, allSettled tolerance, and undecryptable UI exist; multi-device/key recovery is limited.
Watch/Files/Study/Focus Lock	Partial	Core foundations exist; broad two-user, storage, lifecycle, mobile, and production evidence remains.

Foundation	Status	What is preserved
Public homepage boundary	Partial/Unverified	Public room controls are removed; authenticated create/join behavior must remain verified.

Final product target

A user authenticates safely, completes a privacy-aware profile, controls relationships, uses separate Rooms and Personal Chat, collaborates with synchronized Watch/Files/Study/voice/documents, converts eligible rooms into groups, uses Music within official provider limits, and later opts into permission-aware AI. Existing room, E2EE, RLS, lifecycle, realtime, and session foundations remain intact.